

Figure 1:

Se calculează determinantul:

$$\begin{vmatrix} 1 & \cos x_0 & \sin x_0 & \dots & \cos(nx_0) & \sin(nx_0) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & \cos x_{2n} & \sin x_{2n} & \dots & \cos(nx_{2n}) & \sin(nx_{2n}) \end{vmatrix} \\ = 2^{2n^2} \prod_{2n \geq i > j \geq 0} \sin \frac{x_i - x_j}{2} \neq 0$$

dacă $x_i \in [0, 2\pi)$. Ideea: se folosesc relațiile lui Euler, Vandermonde:

$$\begin{aligned} \cos x &= \frac{e^{jx} + e^{-jx}}{2} = \frac{z^2 + 1}{2z}, \sin x = \frac{e^{jx} - e^{-jx}}{2j} = \frac{z^2 - 1}{2jz} \\ z &= e^{jx}, \dots, \cos nx = \frac{z^{2n} + 1}{2z^n}, \sin nx = \frac{z^{2n} - 1}{2jz^n} \end{aligned}$$

Fără a calcula determinantul, folosind schimbarea de variabilă de mai sus avem:

$$\frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) = \frac{P_{2n}(z)}{z^n}$$

unde P_{2n} este un polinom de grad $2n$ care are cel mult $2n$ rădăcini, deci ecuația:

$$\frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) = 0$$

are cel mult $2n$ soluții (definiția T-sistemelor).

$$V(x_0, \dots, x_n) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & x_0 & \dots x_n \\ 1 & x_0^n & \dots x_n^n \end{vmatrix} = \prod_{n \geq i > j \geq 0} (x_i - x_j)$$

(vezi culegere de probleme de analiza si algebra, Donciu si Flondor, vol. I sau culegere de probleme de algebra de Fadeev si Sominski)

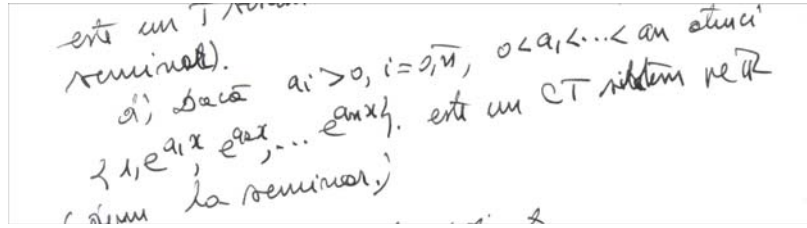


Figure 2:

Calcul $pt. n = 1$:

$$\begin{aligned}
 \begin{vmatrix} 1 & \cos x_0 & \sin x_0 \\ 1 & \cos x_1 & \sin x_1 \\ 1 & \cos x_2 & \sin x_2 \end{vmatrix} & \stackrel{?}{=} 2^2 \sin \frac{x_1 - x_0}{2} \sin \frac{x_2 - x_1}{2} \sin \frac{x_2 - x_0}{2}. \\
 \begin{vmatrix} 1 & \cos x_0 & \sin x_0 \\ 1 & \cos x_1 & \sin x_1 \\ 1 & \cos x_2 & \sin x_2 \end{vmatrix} &= \begin{vmatrix} 0 & \cos x_0 - \cos x_2 & \sin x_0 - \sin x_2 \\ 0 & \cos x_1 - \cos x_2 & \sin x_1 - \sin x_2 \\ 1 & \cos x_2 & \sin x_2 \end{vmatrix} = \\
 &= \begin{vmatrix} 2 \sin \frac{x_2 - x_0}{2} \sin \frac{x_0 + x_2}{2} & -2 \sin \frac{x_2 - x_0}{2} \cos \frac{x_0 + x_2}{2} \\ 2 \sin \frac{x_2 - x_1}{2} \sin \frac{x_1 + x_2}{2} & -2 \sin \frac{x_2 - x_1}{2} \cos \frac{x_1 + x_2}{2} \\ -4 \sin \frac{x_2 - x_0}{2} \sin \frac{x_2 - x_1}{2} & \begin{vmatrix} \sin \frac{x_0 + x_2}{2} & \cos \frac{x_0 + x_2}{2} \\ \sin \frac{x_1 + x_2}{2} & \cos \frac{x_1 + x_2}{2} \end{vmatrix} \end{vmatrix} = \\
 &= -4 \sin \frac{x_2 - x_0}{2} \sin \frac{x_2 - x_1}{2} \sin \left(\frac{x_0 + x_2}{2} - \frac{x_1 + x_2}{2} \right)
 \end{aligned}$$

Ideea $\{1, e^{a_1 x}, \dots, e^{a_n x}\}$ sunt soluții lineare independente ale ec. dif.

$$y^{(n+1)} + \alpha_n y^{(n)} + \dots + \alpha_1 y' = 0 \quad (1)$$

a carei ec. caracteristică:

$$r^{n+1} + \alpha_n r^n + \dots + \alpha_0 r = 0$$

are rădăcinile:

$$\{0 < a_1 < \dots < a_n\}.$$

CF. unei teoreme din "Teoreme de medie din analiza matematică legătura lor cu teoria interpolării" de Elena Popoviciu (pag. 36) Proprietate de a fi CT sistem este echivalent cu Wronskianul $W(1, e^{a_1 x}, \dots, e^{a_k x}) > 0, k = 1, \dots, n$. Dar:

$$W(1, e^{a_1 x}, \dots, e^{a_k x}) = e^{a_1 x} \cdot \dots \cdot e^{a_k x} V(1, \alpha_1, \dots, \alpha_k) > 0.$$

Dem.

$$\begin{aligned}
 \sum_{i=0}^n c_i u_i(x_j) &= 0 + f(x_j), j = \overline{0, n} \\
 \sum_{i=0}^n c_i u_i(x) &= L_n(f; u)(x) + 0
 \end{aligned}$$

1. Dacă $\{u_0, u_1, \dots, u_n\}$ este un CT restul lui f pe Δ este

$$L_n(f; u)(x) = - \frac{\begin{vmatrix} u_0(x_0) & \dots & u_n(x_0) & f(x_0) \\ u_0(x_1) & \dots & u_n(x_1) & f(x_1) \\ \vdots & & \vdots & \vdots \\ u_0(x_n) & \dots & u_n(x_n) & f(x_n) \\ u_0(x) & \dots & u_n(x) & 0 \end{vmatrix}}{\Delta \begin{pmatrix} u_0 & \dots & u_n \\ x_0 & \dots & x_n \end{pmatrix}}$$

Figure 3:

sist. de mai sus cu nec. c_i are sol. unică. Rezultă că minorul caracteristic este nul, adică:

$$\begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) & 0 + f(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) & 0 + f(x_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) & 0 + f(x_n) \\ u_0(x) & u_1(x) & u_2(x) & \dots & u_n(x) & L_n(f; u)(x) + 0 \end{vmatrix} = 0$$

rezultă:

$$\begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) & 0 \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) & 0 \\ u_0(x) & u_1(x) & u_2(x) & \dots & u_n(x) & L_n(f; u)(x) \end{vmatrix} + \begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) & f(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) & f(x_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) & f(x_n) \\ u_0(x) & u_1(x) & u_2(x) & \dots & u_n(x) & 0 \end{vmatrix} = 0$$

Rezultă:

$$L_n(f; u)(x) = - \frac{\begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) & f(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) & f(x_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) & f(x_n) \\ u_0(x) & u_1(x) & u_2(x) & \dots & u_n(x) & 0 \end{vmatrix}}{\begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) \end{vmatrix}} \quad (2)$$

3. Fie $u_0, \dots, u_n$ un C.T. sistem n : $L_n(\frac{1}{2}u)\alpha_j = \sum_{i=0}^n d_i u_i(x)$

Să se arate că $\alpha_n = \frac{\Delta(u_0 u_1 \dots u_{n-1}, f)}{\Delta(u_0 u_1 \dots u_n)}$

(se numește diferență divizată a funcției f raport. cu sistemul $\{u_i\}$).

Figure 4:

Ideea: se determină coef. lui u_n din (*):

$$\alpha_n = - \frac{(-1)^{n+2+n+1} \begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_{n-1}(x_0) & f(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_{n-1}(x_1) & f(x_1) \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_{n-1}(x_n) & f(x_n) \end{vmatrix}}{\begin{vmatrix} u_0(x_0) & u_1(x_0) & u_2(x_0) & \dots & u_n(x_0) \\ u_0(x_1) & u_1(x_1) & u_2(x_1) & \dots & u_n(x_1) \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ u_0(x_n) & u_1(x_n) & u_2(x_n) & \dots & u_n(x_n) \end{vmatrix}}.$$